Practice workbook · Getting one exact trig ratio from another

Using sin^2 + cos^2 = 1 (and tan = sin/cos) to turn one known trig ratio of an angle into the exact value of another, with the right sign.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-02

1. Question 2 · part a · 4 marks

2019 JUN · Paper 01 (regional R)

Worked-example retry · Attempt part a only.

2

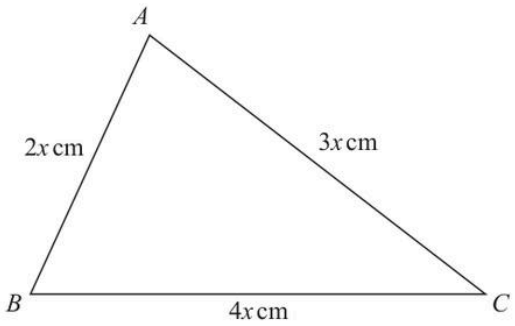


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC in which

$AB = 2x$ cm $\quad AC = 3x$ cm $\quad BC = 4x$ cm

(a) Show that $\sin ABC = \frac{3\sqrt{15}}{16}$

(4)

Given that the area of triangle ABC is $\frac{75\sqrt{15}}{64}$ cm^2

(b) find the value of x.

(2)

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1. Question 2 · part a · 4 marks · Working and answer

Working and answer

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2. Question 2 · part a.i · 0 marks

2021 JUN · Paper 01

Further practice · Attempt part a.i only.

2 Angle α is acute such that $\cos\alpha = \frac{3}{5}$

Angle β is obtuse such that $\sin\beta = \frac{1}{2}$

(a) Find the exact value of

(i) $\tan\alpha$

(ii) $\tan\beta$

(3)

(b) Hence show that

$$\tan(\alpha + \beta) = \frac{m\sqrt{3} - n}{n\sqrt{3} + m}$$

where m and n are positive integers whose values are to be found.

(3)

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2. Question 2 · part a.i · 0 marks · Working and answer

Working and answer

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3. Question 2 · part a.ii · 3 marks

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Further practice · Attempt part a.ii only.

2 Angle α is acute such that $\cos\alpha = \frac{3}{5}$

Angle β is obtuse such that $\sin\beta = \frac{1}{2}$

(a) Find the exact value of

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(b) Hence show that

$$\tan(\alpha + \beta) = \frac{m\sqrt{3} - n}{n\sqrt{3} + m}$$

where m and n are positive integers whose values are to be found.

(3)

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3. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

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4. Question 6 · part a.ii · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part a.ii only.

6 In triangle ABC, $AC = x$ cm, $AB = (x + 3)$ cm and $\angle ABC = 30°$

Given that $\angle ACB = \theta°$ where $0 < \theta < 90$

(a) show that

(i) $\sin\theta° = \dfrac{x + 3}{2x}$

(ii) $\cos\theta° = \dfrac{\sqrt{3x^2 - 6x - 9}}{2x}$

(5)

Given that the size of $\angle BAC$: the size of $\angle ABC = 7 : 2$

(b) find the exact value of x

Give your answer in the form $a + a\sqrt{b}$ where a and b are prime numbers.

(5)

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4. Question 6 · part a.ii · 5 marks · Working and answer

Working and answer

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